

A spectral remainder in the Hellinger-to-information implication

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October 10, 2026

Abstract

We retain a quadratic remainder in the standard implication from the Boolean Hellinger inequality to the Courtade–Kumar information bound. Assuming the Hellinger inequality for a Boolean function f , its information deficit is at least $[\rho^2 - \text{Var}(T_\rho f)]/6$. Combined with the Friedgut–Kalai–Naor theorem, this gives distance to a signed coordinate at most $C\Delta_\rho(f)/[\rho^2(1-\rho^2)]$ for a universal constant C . The general-function statement depends on the full-bias Hellinger inequality asserted in the cited source. For antipodal functions, the same conclusion follows from Ky’s separate Hellinger theorem. This is a potentially new quantitative refinement of an existing argument; publication priority has not been established.

1 Setting and source assumption

Let $n \geq 1$, let X be uniform on $\{-1, 1\}^n$, and let $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$. For $0 \leq \rho \leq 1$, obtain Y_ρ by independently flipping each coordinate of X with probability $(1 - \rho)/2$. The noise operator is

$$T_\rho f(y) = \mathbb{E}[f(X) \mid Y_\rho = y].$$

All expectations use the uniform cube unless otherwise stated. All logarithms are natural, and mutual information is measured in nats. With $0 \log 0 = 0$, write

$$h_2(p) = -p \log p - (1 - p) \log(1 - p), \quad \psi(r) = \log 2 - h_2((1 + r)/2), \quad R(r) = \sqrt{1 - r^2}.$$

The information deficit from a signed coordinate is

$$\Delta_\rho(f) = \psi(\rho) - I(f(X); Y_\rho).$$

For $S \subseteq [n]$, put $\hat{f}(S) = \mathbb{E}[f(X) \prod_{i \in S} X_i]$ and $W_1(f) = \sum_{i=1}^n \hat{f}(\{i\})^2$.

The input to the argument is the Hellinger inequality

$$R(\mathbb{E}f) - \mathbb{E}R(T_\rho f) \leq 1 - R(\rho). \tag{H}$$

Theorem 1.1 of the repository manuscript [1] asserts (H) for all such f , including functions with nonzero mean. We prove an implication from this input and do not give a new proof of the general Hellinger inequality. The qualitative implication to mutual information is due to Anantharam et al. [2].

2 The spectral remainder

Theorem 1. *Let $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ and $0 \leq \rho \leq 1$. If (H) holds for this f and ρ , then*

$$\begin{aligned} \Delta_\rho(f) &\geq \frac{1}{6} \left(\rho^2 - \text{Var}(T_\rho f) \right) \\ &= \frac{\rho^2}{6} \left((\mathbb{E}f)^2 + \sum_{|S| \geq 2} (1 - \rho^{2|S|-2}) \widehat{f}(S)^2 \right). \end{aligned} \quad (1)$$

Consequently,

$$\Delta_\rho(f) \geq \frac{\rho^2(1 - \rho^2)}{6} (1 - W_1(f)). \quad (2)$$

Proof. Define $\phi(t) = h_2((1 + \sqrt{1 - t^2})/2)$ for $0 \leq t \leq 1$. For $0 < t < 1$, putting $v = \sqrt{1 - t^2}$ gives

$$\phi'(t) = \frac{t \operatorname{atanh}(v)}{v}, \quad \phi''(t) = \frac{\operatorname{atanh}(v) - v}{v^3} = \sum_{j \geq 1} \frac{v^{2j-2}}{2j+1} \geq \frac{1}{3}.$$

Moreover $\phi'(t) \rightarrow 0$ as $t \downarrow 0$. It follows that $q(t) = \phi(t) - t^2/6$ is convex and nondecreasing on $[0, 1]$.

Put $g = T_\rho f$, $m = \mathbb{E}f$, $Z = R(g)$, $h = R(m)$, $b = \mathbb{E}Z$, and $s = R(\rho)$. Concavity of R and $\mathbb{E}g = m$ give $0 \leq b \leq h \leq 1$. Assumption (H) gives $h - b \leq 1 - s$. Jensen's inequality and convexity therefore yield

$$q(h) - \mathbb{E}q(Z) \leq q(h) - q(b) \leq q(1) - q(1 - h + b) \leq q(1) - q(s). \quad (3)$$

For the middle inequality, translate the interval $[b, h]$ rightward by $1 - h$; increments of a convex function over an interval of fixed length increase with its left endpoint. The last inequality follows from $1 - h + b \geq s$ and monotonicity of q .

The posterior-mean identity gives $I(f(X); Y_\rho) = \phi(h) - \mathbb{E}\phi(Z)$, and $\psi(\rho) = \phi(1) - \phi(s)$. Substituting $q(t) = \phi(t) - t^2/6$ into (3) gives

$$\Delta_\rho(f) \geq \frac{1}{6} (1 - s^2 - h^2 + \mathbb{E}Z^2) = \frac{1}{6} (\rho^2 + m^2 - \mathbb{E}g^2),$$

which is (1). Parseval and $\widehat{T_\rho f}(S) = \rho^{|S|} \widehat{f}(S)$ give its Fourier expansion. For $|S| \geq 2$, $1 - \rho^{2|S|-2} \geq 1 - \rho^2$; also $1 \geq 1 - \rho^2$ for the mean term. This proves (2). The argument includes the endpoints by continuity. $\square$

Corollary 1 (Stability). *There is a universal $C < \infty$ such that, if (H) holds for f and $0 < \rho < 1$, then*

$$\text{dist}(f, \text{dictators}) := \min_{\substack{1 \leq i \leq n \\ \sigma \in \{-1, 1\}}} \Pr[f(X) \neq \sigma X_i] \leq \frac{C \Delta_\rho(f)}{\rho^2(1 - \rho^2)}.$$

Proof. Set $\eta = 1 - W_1(f)$. The Friedgut–Kalai–Naor theorem [3] gives a Boolean function a depending on at most one coordinate, with $\Pr[f \neq a] \leq K \sum_{|S| \geq 2} \widehat{f}(S)^2 \leq K\eta$, for some universal $K \geq 1$. If $\eta < \min(1/4, 1/(8K))$, then $(\mathbb{E}f)^2 \leq \eta$ implies $|\mathbb{E}f| < 1/2$. Every constant consequently has disagreement greater than $1/4$ with f , whereas the FKN approximant has disagreement less than $1/8$. Thus a is a signed coordinate. For larger η , the trivial bound $\text{dist}(f, \text{dictators}) \leq 1/2$ gives the same conclusion $\text{dist}(f, \text{dictators}) \leq C_0\eta$ after enlarging the universal constant. Apply (2) with $C = 6C_0$. $\square$

Corollary 2 (Antipodal functions). *If $f(-x) = -f(x)$ for every x , then the conclusions of Theorem 1 and Corollary 1 follow from Ky’s antipodal Hellinger theorem [5, Theorem 1.1], without using the general full-bias input of [1].*

Proof. Antipodality gives $\mathbb{E}f = 0$. Ky’s theorem states $\mathbb{E}\sqrt{1 - (T_\rho f)^2} \geq \sqrt{1 - \rho^2}$, which is exactly (H) when $\mathbb{E}f = 0$. Apply the preceding results. $\square$

3 Comparison and scope

Ky and Tran [4, Theorem 1.1] prove

$$\text{dist}(f, \text{dictators}) \leq \frac{10^8 \Delta_\rho(f)}{\rho^2(1 - \rho)h_2((1 - \rho)/2)}.$$

For increasing functions they remove $1 - \rho$ from that denominator. Their Section 5.6 leaves open whether this extra factor can be removed for general functions. Conditional on (H), our denominator has order $1 - \rho$ as $\rho \uparrow 1$, improving the general bound’s order $(1 - \rho)^2 \log(1/(1 - \rho))$. Our unspecified universal constant means this is an asymptotic comparison, not a pointwise comparison of numerical constants. We do not achieve the increasing-function order $(1 - \rho) \log(1/(1 - \rho))$ or resolve the sharp general question.

Dimension-independent stability is already known. The potential contribution here is the explicit spectral remainder and the resulting noise dependence. The convex-profile transfer is already in [2] and is reproduced in [1]. Our extra step is to subtract $t^2/6$ before that transfer. This coefficient is the largest allowed by this global quadratic-subtraction argument, since $\phi''(t) \rightarrow 1/3$ as $t \uparrow 1$. We do not claim that $1/6$ is the optimal coefficient in (1).

The separate antipodal input [5] was posted on October 7, 2026. Its Section 6.2 discusses quantitative stability as a further question. Corollary 2 supplies an information-stability consequence; it does not quantify all losses in that paper’s matrix argument or give its equality classification for general product channels. A targeted literature check has not established priority for the present refinement. This manuscript is a research note, not a claim of a new construction or a new proof of the general Hellinger inequality.

4 Checks and disclosure

The accompanying script `check_hellinger_stability.py` enumerates every Boolean function in dimensions one through four at eight correlations. All 526,496 cases passed checks of the Hellinger input, the spectral remainder, and the Fourier identity, at floating-point tolerance 10^{-12} . These finite diagnostics are not a proof.

The source’s complete `verification/verify_all.py` workflow was also rerun on October 7, 2026 with Python 3.11.14, SymPy 1.14.0 and `clang++`; all supplied arithmetic-certificate stages passed. The accompanying `source-verification-summary.json` records the source revision, file hashes, and outcomes. This checks finite certificates, not the entire analytic proof of [1]. The note, diagnostic code, and internal reviews were prepared with assistance from Codex. No Lean kernel verification or external peer review is claimed.

References

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